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Assigned System: Database Query Performance Profiler

ARCHITECTURAL JUSTIFICATION DOCUMENT

1. Architecture Selection

We chose **Layered Architecture** for the Database Query Performance Profiler system.

2. Two Scenario-Related Reasons

1. **High Ingestion Throughput:** The profiler must ingest up to 5,000 log records per minute from PostgreSQL and MySQL hosts. Layered architecture keeps log ingestion, execution plan parsing, and metric storage within the same process memory space, avoiding inter-service HTTP/RPC network latency.
2. **Clear Separation of System Modules:** The core business logic (EXPLAIN plan parsing, candidate index generation, and weekly digest compilation) is separated from the presentation layer (DBA dashboard). Updates to parsing algorithms or log formats can be made without impacting user views or persistent storage logic.

3. Security Advantage

NFR-002 mandates that literal values (such as sensitive IDs or email addresses in SQL WHERE clauses) must be redacted before storage or display. In a Layered Architecture, the business logic layer enforces a strict security boundary. Incoming raw query strings pass through parameterization and redaction algorithms in the **Log Ingestion Component** before entering the **Database Component** or **User Interface Component**, ensuring raw unredacted data cannot leak into storage or UI layers.

4. Performance Benefit (Meeting Performance Requirements)

Processing 5,000 log entries per minute requires an average execution budget under 12 milliseconds per record. Layered Architecture allows direct in-memory method invocations between the **Log Ingestion**, **Plan Parser**, and **Database** components. Eliminating network serialization and remote API overhead ensures high-volume log parsing and storage completes well within the polling window without causing performance degradation on the host database.